

ECS 289D Seminar in OS: Datacenter Systems for LLMs

Yang Zhou
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With slides adapted from Prof. Amanda Raybuck last-year offering

About me

- Yang Zhou
 - BS in CS from Peking University, China in 2018
 - PhD in CS from Harvard University, USA in 2024
 - Postdoc from UC Berkeley Sky Computing lab in 2025
 - First-year assistant professor at UC Davis
- Research focus:
 - Equal interests in core systems and ML systems research,
 - e.g., efficient LLMs, GPU communication, heterogeneous computing.
 - Currently working on UCCL for GPU communication
 - <https://github.com/uccl-project/uccl>

Agenda for today

Introduction to datacenter systems for LLMs

Course logistics and more

Topic overview

How to give a good talk

Introduction to datacenter systems for LLMs



LLM Booming: Companies & Models

- LLM companies are rapidly emerging
- Many new LLM models are being developed
- The LLM landscape includes diverse offerings
- Interest in large language models is increasing

LLM Use Cases

- LLMs power intelligent chatbots and virtual assistants
- They enable advanced content generation and summarization
- LLMs are used for complex data analysis and extraction
- They facilitate code generation and debugging assistance



xAI Colossus for LLMs: 300MW, 200k GPUs



Colossus: Total GPUs: **200,000**

Phase 1: **122 days** - 100k GPUs fully training synchronously. From scratch.



Phase 2: **92 days** to expand to 200K GPUs

xAI Colossus 2 for LLMs: 1.1GW, 550k GPUs



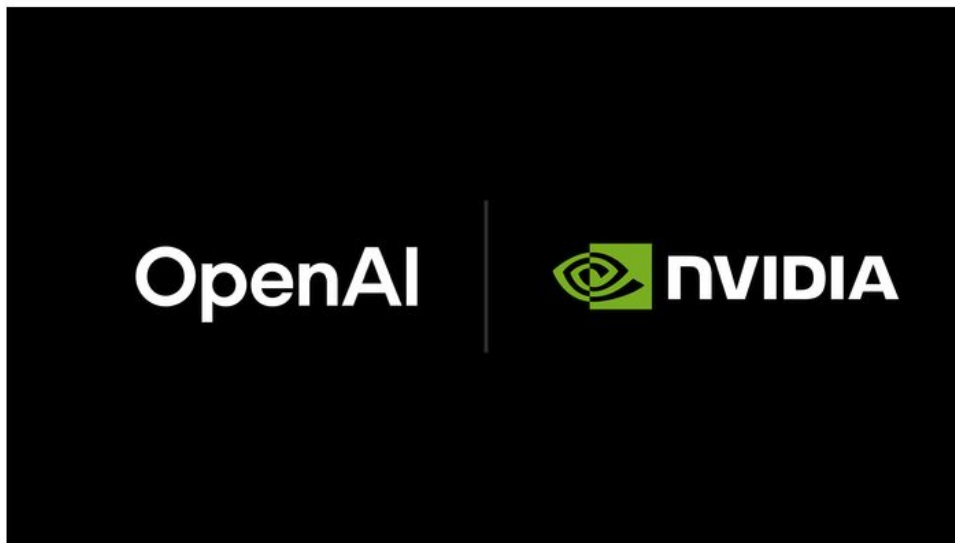
March 2025



August 2025

OpenAI and NVIDIA Announce Strategic Partnership to Deploy 10 Gigawatts of NVIDIA Systems

September 22, 2025



News

- > Strategic partnership enables OpenAI to build and deploy at least 10 gigawatts of AI data centers with NVIDIA systems representing millions of GPUs for OpenAI's next-generation AI infrastructure.
- > To support the partnership, NVIDIA intends to invest up to \$100 billion in OpenAI progressively as each gigawatt is deployed.
- > The first gigawatt of NVIDIA systems will be deployed in the second half of 2026 on the NVIDIA Vera Rubin platform.

What is a datacenter?

- Large facility housing primarily commodity computers
 - 10,000s - millions of machines
 - Commodity servers (e.g., Dell, Open Compute Project (OCP))
- Interconnected by a commodity network
 - Ethernet
- Commodity?
 - Cheap! Reduces operating costs (vs. custom parts)
 - “One-size-fits-all” components
 - All good as long as they improve (Moore’s Law)
 - Strong drive by DC operators to commoditize all infrastructure

Why Datacenters?

- Consolidation
 - Run many people's workloads on the same infrastructure
 - Use infrastructure more efficiently (higher utilization)
 - Leverage workload synergies (e.g., caching)
- Virtualization
 - Build your own (virtual) private infrastructure quickly and cheaply
 - Move it around + scale it up/down anywhere, anytime
- Outsourcing
 - No need to maintain an on-premise set of servers
 - Expertise is provided by the datacenter vendor

Datacenters Today

- Over 8000 datacenters globally
- Over 2600 datacenters in the US
- Huge energy consumers – almost 2% of global energy use
 - Usually built near energy sources (hydroelectric, wind, solar)



Google datacenters



Amazon AWS datacenters

Why researching datacenters systems for LLMs?

- Datacenters have become the cornerstone of the LLM workloads
 - The world's most valuable companies are datacenter operators
- They comprise the state-of-the-art in computing
 - Scale, technology, applications
- They combine networking, systems, and economic lessons
 - Increasingly important knowledge
- They are rarely discussed in class
 - Fairly recent topic
 - Crosses disciplines

Course logistics and more

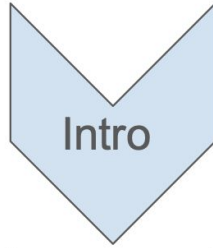
Course logistics

- Graduate-level, seminar-based, research-focused course
- The goals of the course are:
 - To learn about classic and cutting-edge datacenter systems for LLMs
 - To practice reading and discussing research/technical papers
 - To conduct a research project
- Three main components:
 - Reading list + paper reviews + discussion
 - Leading a lecture
 - Research project

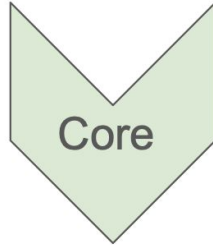
Assumptions

- Programming in C/C++ and Python
- Basic OS and Networking concepts
 - Virtual address, isolation
- Basic knowledge about computer architecture
 - GPU, registers, DMA engines

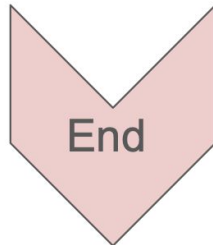
Schedule



- Class Introduction
- Define projects and find groups
- Decide on paper(s) to present



- Research paper presentations & discussions
- Work on project



- Present project
- Submit project and source code

Paper Readings

- Most of our material is not (or insufficiently) covered in textbooks
- Readings will be **research papers or famous tech blogs**
 - Seminal papers in the field, research papers that show what is possible
 - Details of key ideas in datacenter systems for LLMs
- We will read & discuss 2-4 papers per week
 - Each session will cover
 - 1 required reading, 1-2 optional readings
 - 5% bonus points if you also write (good) reviews for optional readings
 - 2 presentations + discussions, 40 min each
 - 1 presentation for required reading, 1 for optional readings
 - Presentations and discussions can blend together

Paper Reviews: What you need to do

- Read the paper before class
- Write a short review that includes:
 - A short (3-4 sentence) summary of the paper
 - Some advantages or disadvantages of the approach
 - Any other questions or comments you have
- Submit on canvas by **11:59pm the night before class**
 - See canvas assignments

Paper Presentations: What you need to do

- Select a paper to present
 - First-come, first-served, due by **11:59 pm Tue 9/30**
 - Select a paper via Canvas comment under “Paper Assignments” post
 - Available today noon
 - If everything is taken, up to 2 students may present 1 paper
- When it is your time, prepare and present the paper for discussion
 - See schedule for order of papers
 - Send me a copy of your slides after the session, so I can put on the course website
- Participate in discussion!
 - Worth 10% of your grade

Projects: What you need to do

- Teams of 3-5 students work on each project
- Define your project
 - Whatever related to systems and LLMs
 - Talk with me on the compute resources you need
- Prepare a proposal
 - Paper introduction defining the project
 - Evaluation plan and timeline/milestones
- **Start working!**
 - Check in with me regularly (mandatory mid-quarter check in by 11/06)

Project Discussion

- Post (and answer) questions on Canvas
 - Gets you better, faster, collaborative answers
- Check in with me regularly
 - Use office hours
 - 1 mandatory mid-quarter check in (11/06)

Grading

Straight grading scale ($\geq 90\%$ A, 87-89 B+, 83-86 B, 80-82 B-, etc.)

- Project: 50%
 - 1% project proposal
 - 4% mid-term report
 - 10% final project presentation
 - 35% final report and code
- Class presentation: 25%
- Paper reviews: 15%
- Class participation: 10%
 - Class attendance, in-class discussions, and online discussions

Academic Integrity and Generative AI Policy

- Talking with others in the class about papers, projects is OK, encouraged
- Give credit to any resources you borrow for presentations, write-ups
 - E.g., if you use slides from a conference presentation in your paper, give credit
 - When in doubt, cite it!
- I am willing to allow the use of Generative AI, but within reason (& ACK use)
 - Using it for proposal/project writing:
 - You may use it for a “first draft” but I expect human-made revisions over what the model produces
 - OK to use for grammar and clarity improvements
 - Using it for code generation:
 - OK , but acknowledge use in write-ups
 - You (and your group) are responsible for understanding anything you turn in!

Topic overview

Four main topics

- Datacenter networking
- Host networking
- LLM Inference
- LLM Training

Datacenter networking

[The Tail at Scale](#)

optional [Attack of the Killer Microseconds](#)

[A Scalable, Commodity Data Center Network Architecture](#)

optional [VL2: A Scalable and Flexible Data Center Network](#)

[Data Center TCP \(DCTCP\)](#)

optional [Swift: Delay is Simple and Effective for Congestion Control in the Datacenter](#)

[Design Guidelines for High Performance RDMA Systems](#)

optional [Deconstructing RDMA-enabled Distributed Transactions: Hybrid is Better!](#)

Host networking

[RDMA over Ethernet for Distributed AI Training at Meta Scale](#)

optional [An Extensible Software Transport Layer for GPU Networking](#)

[IX: A Protected Dataplane Operating System for High Throughput and Low Latency](#)

optional [Arrakis: The Operating System is the Control Plane](#)

[Shenango: Achieving High CPU Efficiency for Latency-sensitive Datacenter Workloads](#)

optional [Snap: a Microkernel Approach to Host Networking](#)

[Demystifying NCCL: An In-depth Analysis of GPU Communication Protocols and Algorithms](#)

optional [MSCCL++: Rethinking GPU Communication Abstractions for Cutting-Edge AI Applications](#)

LLM Inference

[Efficient Memory Management for Large Language Model Serving with PagedAttention](#)

optional [vAttention: Dynamic Memory Management for Serving LLMs without PagedAttention](#)

[DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving](#)

optional [Optimizing SLO-oriented LLM Serving with PD-Multiplexing](#)

[FlashInfer: Efficient and Customizable Attention Engine for LLM Inference Serving](#)

optional [XGrammar: Flexible and Efficient Structured Generation Engine for Large Language Models](#)

[NanoFlow: Towards Optimal Large Language Model Serving Throughput](#)

optional [Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve](#)

LLM Training

[FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness](#)

optional [FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning](#)

optional [FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision](#)

ZeRO: Memory Optimizations Toward Training Trillion Parameter Models

optional [PyTorch FSDP: Experiences on Scaling Fully Sharded Data Parallel](#)

optional [Everything about Distributed Training and Efficient Finetuning](#)

[Insights into DeepSeek-V3: Scaling Challenges and Reflections on Hardware for AI Architectures](#)

optional [DeepSeek Open Infra](#)

[Gemini: Fast Failure Recovery in Distributed Training with In-Memory Checkpoints](#)

optional [Alpa: Automating Inter- and Intra-Operator Parallelism for Distributed Deep Learning](#)

How to give a good talk

Slides and advice adapted from Thorsten Hoefler
(ETH Zurich) and Simon Peter (UWashington)

Talking about research

- A good researcher can express their knowledge well
 - This assumes we have that knowledge!
 - So only advance to this step once you understand the paper :)
- Why is talking about research useful?
 - Order your thoughts, think about how to explain them
 - Communication with other researchers (your classmates!)
 - Gather feedback
 - Establish relationships
 - Eventually build a career

Why do you care?

- Presenting will be important for your career!
 - This is a good way to convince people to:
 - Give you good grades
 - Give you a job/promotion
 - Give you money/resources
 - Think you're smart
 - Like you, recommend you
- But presentation skills are hard to acquire
 - No one is good at this right away
 - Practice practice practice!

A good research talk

- Is centered around the audience (not you)
 - **Teaches**, engages, provokes, excites listeners
- Provides intuitions to the audience
 - Take away messages, surprises, “wow” effects
- It does not need to
 - Tell every detail (not possible anyway in the time limit)
 - Show off how smart you are

Tip: focus on clearly defined goals

- Pick your goals carefully
 - What do you want to communicate? What should people remember?

Anatomy of a talk

- Motivation, placement
 - 20%
- Key ideas
 - 70-80%
- Evaluation results
 - 0-10%
- Do not present results without an explanation
 - Don't just say "the authors made Application A run 50% faster"
 - Instead say "the authors present the FOO method relying on intuition BAR and that achieves 50% improvement for Application A"

The beginning of your talk

- You have about 2 minutes before your audience dozes off or starts reading email
 - Use them! Make every second count
 - Good approach: Present an abstract of your talk (or an elevator pitch):
 - Problem / motivation
 - Approach / idea
 - Experiments / results
 - Broader meaning / impact
- Answer these questions within two minutes:
 - What is the problem?
 - Why is this talk interesting? Why should I listen?

Communicating the key idea

- Pick a goal for the talk
 - Plan and make key points in your head. Organize the whole talk around these key points. Pick no more than three (better: one)
 - Be explicit, very explicit: “If you remember nothing else from this talk, remember this”
 - Repeat, repeat (but don’t be annoying)
- Do NOT be shallow, be deep
 - Avoid overviews
 - Do NOT ramble
 - Get to the meat quickly (assume we’ve all read the paper already)

Examples

- Are your main weapon
- Make your own examples, try to avoid just using the paper's
- Ideally have a motivating example at the beginning
 - Maybe pose a question to get the audience thinking:
 - e.g., "What is the maximum speedup if we can solve this problem?"
- Illustrate the idea in action
 - From different perspectives
 - Show corner cases, highlight shortcomings
- Images say more than 1000 words!

What to use

- Enthusiasm
 - Be excited, pull the audience with you
 - Don't be afraid to move around, but don't pace or fidget
- Your brain!
 - Review/polish your slides before the talk
 - Have the storyline down
 - Focus on the key ideas
- Animations and graphics
 - Can be helpful (aesthetically as well as informative)
 - Nice slides make people more receptive

What to omit

- Do not present excessive related work
 - You can mention it in your slides, or have backup slides
 - Give credit
- Do not present too many technicalities
 - Audience probably won't follow
 - Put details in the backup slides in case you do get a question on them
- Do not exaggerate with animations
 - Animations are good, but too many are difficult to follow
- Do not clutter your slides

How to present

- Be (or at least appear) confident
 - Don't forget to breathe! :)
- Make eye contact
 - Look around, don't stare at a single person
 - Tip: identify a nodder (these people always exist). They'll give you confirmation
- Watch the audience
 - Sometimes they ask questions, don't let them interrupt you but serve their questions
 - Questions are great, ask some and answer them!
- Finish on time!
 - Skip slides if necessary, never ask "should I continue?"
 - No polite person would ever say, "no, thanks"

Miscellaneous

- Standard stuff
 - Avoid errors no slides
 - Face the audience
 - Check your laptop before
- Practice, practice, practice
 - Give your talk a couple of times before you present it publicly
 - Present to your group mates, me during office hours, your friends, your cat...
- You'll attend more talks here than you'll give
 - Engage, help your fellow students!
 - Ask questions, participate in the discussion, be awake

Miscellaneous

- You may borrow slides
 - But, be aware of the context of the slides
 - E.g., conference slides are intended for a very specialized audience
 - May have to adapt the slides for a more general audience (and the time frame)
 - And give credit
- Timing and cadence
 - Between 1-2 minutes per slide is common
 - Try not to rush too quickly through slides or spend too much time on a slide

Some example (great) presentations

[OSDI '14 - IX: A Protected Dataplane Operating System for High Throughput and Low Latency](#) (first 3min)

[OSDI '14 - Arrakis: The Operating System is the Control Plane](#) (first 3min)

[Building a Better Web Browser - James Mickens - Harvard CS Colloquium 2015](#)
(first 5min)

Next Tuesday, 9/30
Paper Presentation Selection Due